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Battery module and assembling method of the same

Abstract

The present disclosure relates to a battery module and an assembling method of the same including: a cell assembly including a plurality of battery cells and a busbar assembly for electrically connecting the plurality of battery cells; a case forming a cell accommodating space for accommodating the cell assembly therein; and a filler portion provided between the plurality of battery cells and the busbar assembly inside the case and including a flame-retardant material.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION
(1) The present application claims priority under 35 U.S.C. § 119(a) to Korean patent application number 10-2023-0009650 filed on Jan. 25, 2023, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference herein.
BACKGROUND OF THE INVENTION
1. Field

(2) The present disclosure relates to a battery module, which is a collection of secondary batteries or battery cells, and a method of assembling the battery module. More specifically, it relates to a battery module that delays thermal propagation of battery cells and a method of assembling the battery module.

2. Description of the Related Art

(3) Recently, due to fire or explosion accidents occurring during the use of lithium secondary batteries, social concerns about the safety of battery use are increasing. Based on these social concerns, one of the recent major developmental tasks of lithium secondary batteries is to eliminate instabilities such as fire and explosion caused by thermal runaway of battery cells.

(4) In particular, a typical battery module only considers a gap between a tab of any one battery cell and a tab of another adjacent battery cell, but does not consider the height of battery cells. In this case, there is a problem that it is impossible to prevent a gas generated from a battery cell in which thermal runaway has occurred from escaping through an empty space in a tab direction.

SUMMARY OF THE INVENTION

(5) First, a technical problem that one aspect of the present disclosure is intended to solve is to prevent or mitigate escape of a high-temperature gas generated from a battery cell in which thermal runaway has occurred, among one or more battery cells provided inside a battery module, in a tab direction of the battery cell.

(6) Second, a technical problem that another aspect of the present disclosure is intended to solve is to vente a high-temperature gas generated from a battery cell in which thermal runaway has occurred, in an intended path.

(7) Third, a technical problem that still another aspect of the present disclosure is intended to solve is to add a process of positioning a filler portion between tabs of a busbar assembly and a battery to an existing battery module assembly process.

(8) Meanwhile, a battery module according to the present disclosure can be widely applied in the field of green technology, such as electric vehicles, battery charging stations, energy storage systems (ESS), solar power generation (photovoltaics), and wind power generation using batteries. In addition, a battery module according to the present disclosure can be used in eco-friendly mobility including electric vehicles and hybrid vehicles to prevent climate change by suppressing air pollution and greenhouse gas emissions.

(9) To solve the above-described problem, in a battery module according to the present disclosure, a filler made of a flame-retardant and heat-resistant material is positioned in an empty space of a battery module, specifically between a battery cell and a busbar assembly.

(10) More specifically, a battery module according to the present disclosure may include: a cell assembly including a plurality of battery cells and a busbar assembly for electrically connecting the plurality of battery cells; a case defining a cell accommodating space for accommodating the cell assembly therein; and a filler portion provided between the plurality of battery cells and the busbar assembly inside the case and including a flame-retardant material.

(11) The plurality of battery cells may each include: a cell body portion generating or storing electrical energy; and a tab portion protruding from the cell body portion toward the busbar assembly and electrically connecting the cell body portion and the busbar assembly, wherein the filler portion may be positioned in a filler space defined between one side face of the cell body portion from which the tab portion protrudes and the busbar assembly.

(12) The tab portion may include a first tab and a second tab protruding from the cell body portion toward the busbar assembly any one direction among directions perpendicular to the height direction of the case, wherein the busbar assembly may include: a first busbar electrically connected to the first tab and a first busbar frame supporting the first busbar; and a second busbar electrically connected to the second tab and a second busbar frame supporting the second busbar, and wherein the filler space may include: a first filler space provided between a part in which the first tab is positioned and the first busbar frame; and a second filler space provided between a part

in which the second tab is positioned and the second busbar frame.
(13) Along a direction from the first filler space to the second filler space, the length of the first filler space may be different from the length of the second filler space.
(14) The battery module may further include: a heat dissipating portion positioned between the cell assembly and a bottom surface of the cell accommodating space to dissipate heat generated from the plurality of battery cells to the outside through the case, wherein the heat dissipating portion may be positioned between the first filler space and the second filler space.
(15) The filler portion may be formed of a polymer material that is cured after being injected into the filler space.
(16) The filler portion may be filled in the filler space in a liquid form.
(17) The filler portion may include: a body portion including an insertion groove extending along the height direction of the case to insert the tab portion and electrically connect the tab portion to the busbar assembly; and a connecting portion extending from one side of the body portion in a direction toward the busbar assembly.
(18) The battery module may further include: a first insulation cover positioned in a direction from the outside of the first busbar frame to one side face of the case which is positioned closer to the first busbar frame, among both side faces facing the first busbar frame, to insulate between the first busbar and the case; and a second insulation cover positioned in a direction from the second busbar frame to the other face of the case to insulate between the second busbar and the case.
(19) The cell assembly may further include a first end plate and a second end plate respectively connected to the first insulation cover and the second insulation cover to surround the plurality of battery cells.
(20) The battery module may further include a module cover coupled to the first insulation cover, the second insulation cover, the first end plate, and the second end plate.
(21) Based on a bottom face of the cell accommodating space, the maximum height of the filler portion may be more than half of the maximum height of the plurality of battery cells positioned in the cell accommodating space.
(22) Meanwhile, the case may include: a module body including an opening portion in an upper portion thereof, extending along a first direction, which is any one direction among directions perpendicular to the height direction of the case, and formed in a U-shape of which both side faces perpendicular to the first direction are open; and a module cover coupled to the module body to close the opening portion.
(23) The battery module may further include a heat dissipating portion positioned between the cell assembly and a bottom face of the cell accommodating space to dissipate heat generated from the plurality of battery cells to the outside through the case.
(24) The filler portion may further include a heat-resistant material.
(25) Meanwhile, a method of assembling a battery module according to the present disclosure, which is a method of assembling a battery module including a cell assembly including a plurality of battery cells and a busbar assembly for electrically connecting the plurality of battery cells; a case forming a cell accommodating space for accommodating the cell assembly therein, may include: a step of coupling a module cover coupled with one face of the case with the cell assembly; a step of inverting the cell assembly coupled with the module cover; and a step of forming a filler portion between the busbar assembly and the one or more battery cell.
(26) In the step of forming a filler portion, the filler portion may be moved toward the inverted cell assembly to face an inner side of the bus bar assembly.
(27) In the step of forming a filler portion, the filler portion in a liquid state may be injected to be filled between the plurality of battery cells and the busbar assembly.
(28) The method of assembling a battery module may further include a step of coupling a module body defining the cell accommodating space together with the module cover with the module cover, after forming of the filler portion.

(29) In addition, the method of assembling the battery module may further include a step of re-inverting the module body after coupling the module body to the module cover.
(30) First, according to one embodiment of the present disclosure, escape of a gas generated from a battery cell in which thermal runaway has occurred, among a plurality of battery cells provided inside a battery module, in a tab direction of the battery cell may be prevented or mitigated. Through this, the stability of the battery module may be improved.
(31) Second, according to another embodiment of the present disclosure, a high-temperature gas generated from a battery cell in which thermal runaway has occurred may be vented in an intended path.
(32) Third, according to still another embodiment of the present disclosure, a process of positioning a filler portion may be added to an existing battery module assembly process without making a significant change.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows an exploded view of a battery module.
(2) FIG. **2**A shows an example of a cross-section of the battery module; FIG. **2**B shows an enlarged diagram of a part of the cross-section of FIG. **2**A; and FIG. **2**C shows an example of a cross-section of the battery module viewed from another direction.
(3) FIGS. **3**A and **3**B show another example of a filler portion.
(4) FIG. **4**A shows an example of a filler portion molded into a preset shape; and FIG. **4**B shows an enlarged diagram of a part of the filler portion.
(5) FIG. **5** shows an example of a method of injecting a filler portion injected in a liquid form.
(6) FIGS. **6**A to **6**F briefly show an example of a method of assembling the battery module.
(7) FIG. **7** shows a flowchart of an example of the method of assembling the battery module.
DETAILED DESCRIPTION
(8) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the attached drawings. The configuration or control method of the device described below is only for explaining the embodiments of the present disclosure and is not intended to limit the scope of the present disclosure, and the same reference numerals used throughout the specification indicate the same components.
(9) Specific terms used in the present specification are merely for convenience of explanation and are not used to limit the illustrated embodiments.
(10) A battery cell described in the present specification refers to a basic unit of a lithium secondary battery, specifically a lithium-ion battery, that can be used by charging and discharging electrical energy. The main components of the battery cell are a cathode, an anode, a separator, and an electrolyte, and the battery cell is manufactured by putting these main components into a case (or pouch). The battery cell may include a tab that is each connected to the cathode and the anode for electrical connection to the outside and protrudes to the outside of the pouch.
(11) Meanwhile, a battery module described in the present specification refers to a battery assembly in which the battery cells are bundled in one or more numbers and put into a case to protect them from external shock, heat, vibration, or the like. The plurality of battery cells may supply electricity to the outside or receive electricity from the outside through a busbar assembly that is electrically connected.
(12) In addition, a battery pack refers to a set in which a preset number of battery modules are gathered together and electrically connected to achieve a final desired voltage or power.
(13) FIG. **1** is an exploded view of a battery module **200**. Referring to FIG. **1**, the battery module **200** may include a plurality of battery cells **110** and a busbar assembly **150** electrically connected to

the plurality of battery cells **110**. In particular, after the plurality of battery cells **110** are stacked, they may be electrically connected by the busbar assembly **150**. This is referred to as a cell assembly **100**. The busbar assembly **150** may be electrically connected to the outside to store (or charge) electric energy in the plurality of battery cells **110**, or supply the electric energy stored in the plurality of battery cells **110** to the outside (or discharge).
(14) The battery module **200** may further include a case **210** defining a cell accommodating space **280** for accommodating the cell assembly **100**. The case **210** is intended to protect the cell assembly **100** from the outside.
(15) The case **210** may include a module body **219** accommodating the cell assembly **100** and a module cover **211** coupled to the module body **219** to define the cell accommodating space **280**.
(16) The module body **219** may be provided in a channel shape or a U-shape in which an upper portion of the cell accommodating space **280** is opened. Therefore, among the side faces of the module body **219**, both side faces **2197**, **2198** facing each other may also be opened. For example, referring to FIG. **1**, among the side faces of the module body **219**, both side faces extending along the X-direction may be open.
(17) In the present disclosure, the height direction of a case **210** was set to the +Z direction. The +Z direction is a direction assuming that a module cover **211** is at an upper portion a module body **219**. On the other hand, the −Z direction is a direction assuming that the case **210** is turned over and the module cover **211** is positioned at a lower portion of the module body **219**.
(18) In addition, in the present disclosure, the X-direction is one direction among directions perpendicular to the height direction of the case **210**, and when the case **210** has a hexahedral shape, among four side faces forming side faces of the case, two side faces facing each other will extend along the X-direction, and the other two facing side face **2191**, **2192** of among the four side faces will extend along the Y-direction. In the present specification, for convenience, a direction parallel to a long edge of the case was set as the Y-direction, and a direction parallel to a short edge of the case was set as the X-direction.
(19) The cell assembly **100** may further include a buffering member **117** provided between at least a part of the plurality of battery cells **110**. Therefore, the cell assembly **100** may include one or more of the buffer members **117**. The buffering member **117** may be positioned between the battery cells **110**, or after the plurality of battery cells **110** are grouped into a preset number, it may be positioned between the grouped battery cells. The buffer member **117** is intended to reduce the pressure exerted on other battery cells **110** when the battery cell **110** is swelling. Therefore, the cell assembly **100** may include one or more buffering members **117**.
(20) The plurality of battery cells **110** and the buffering member **117** may be provided at preset positions and stacked. For example, referring to FIG. **1**, an example is shown in which long edges of the plurality of battery cells **110** are arranged side by side in the X-direction. Therefore, the plurality of battery cells **110** and the buffering member **117** will be positioned to overlap in the X-direction.
(21) The cell assembly **100** may further include end portions **180** at both ends of the cell assembly **100**, that is, when the plurality of battery cells **110** are stacked, on exposed faces of the battery cells **110** positioned at the first and last ends.
(22) The end portion **180** is intended to prevent both side faces of the cell assembly **100** from being exposed to the outside.
(23) The battery module **200** may further include an insulating cover **170** coupled to both side faces among the side faces the cell assembly except for the side faces where the end portion **180** is positioned. That is, the end portion **180** and the insulating cover **170** may be connected perpendicularly to each other.
(24) The insulating cover **170** will extend side by side to the busbar assembly **150**, and both ends of the insulating cover **170** will be connected to the end portion **180**.
(25) Meanwhile, each of the plurality of battery cells **110** may include a rectangular parallelepiped-

shaped cell body portion **115** (see FIG. **2**C) that produces or stores electrical energy; and a tab portion **119** (see FIG. **2**C) protruding from the cell body portion **115** along the X-direction (first direction).

(26) Specifically, the tab portion **119** may include: a first tab **119***a* (see FIG. **2**C) and a second tab **119***b* (see FIG. **2**C) protruding from both side faces of the cell body portion **115** in a direction away from the cell body portion **115**. This is an example, and both of the tab portions **119** may be provided on a single side face.

(27) Referring to FIGS. **1** and **2**C, the cell assembly **100** may include a busbar assembly **150** into which the first tab **119***a* and the second tab **119***b* are each inserted and electrically connected. Specifically, the busbar assembly **150** may include: a first busbar **1511** (see FIG. **2**A) electrically connected to the first tab **119***a* and a first busbar frame **1551** supporting the first busbar **1511** (see FIG. **2**A); and a second busbar **1512** (see FIG. **2**A) electrically connected to the second tab **119***b* and a second busbar frame **1552** supporting the second busbar **1512**.

(28) In other words, since the first tab **119***a* and the second tab **119***b* are disposed on both sides of the cell body portion **115**, corresponding thereto, the first busbar **1511** and the second busbar **1512**, which are electrically connected to the first tab **119***a* and the second tab **119***b*, respectively, will also be positioned on the side faces on which the first tab **119***a* and the second tab **119***b* are disposed. Likewise, the first busbar frame **1551** and the second busbar frame **1552** may also be positioned on the side face on which the first tab **119***a* and the second tab **119***b* are positioned.

(29) The busbar assembly **150** may further include a frame connecting portion **155** connecting the first busbar frame **1551** and the second busbar frame **1552** to each other and supporting the same. The cell assembly **100** may further include a wire (not shown) provided on the frame connecting portion **155** to electrically connect the busbar assembly **150** to the outside.

(30) In addition, the end portion **180** may be provided with a first end plate **181** and a second end plate **182** coupled to both side faces of the cell assembly, among the side faces of the cell assembly, which are provided side by side in a direction from the first tab **119***a* toward the second tab **119***b*.

(31) Therefore, the plurality of battery cells **110** and the buffering member **117** will be positioned between the first end plate **181** and the second end plate **182**.

(32) When the insulating cover **170** is coupled to the first end plate **181** and the second end plate **182**, it may further include: a first insulating cover **171** provided outside the first busbar **1551** and the first busbar frame **1551**; and a second insulating cover **172** provided outside the second busbar **1512** and the second busbar frame **1552**.

(33) The first end plate **181**, the second end plate **182**, the first insulating cover **171**, and the second insulating cover **172** will have a form including the one or more battery cells **110** therein.

(34) Meanwhile, the case **210** may further include: a base panel forming a bottom face of the cell accommodating space **280**, that is, a lower face of the module body **219**; and a first side panel **2191** and a second side panel **2192** bent at both edges provided in parallel with the Y-direction, among the edges of the base panel **2194**, to extend toward the module cover **211**. Through this, the module body **219** may be in the form of a U-shaped channel in which both faces along the X-direction are open and the upper face is open.

(35) The battery module **200** may further include a heat dissipating portion **295** provided between the module body **219** and the cell assembly **100** to be in contact with the cell assembly **100**. The heat dissipating portion **295** is intended to radiate heat that is generated from the cell assembly **100** to the outside. For example, the heat dissipating portion **295** may be provided in the form of a heat dissipating pad or thermal adhesive with excellent thermal conductivity.

(36) FIG. **2**A shows an example of a cross-section of the battery module. That is, FIG. **2**A is an example of a cross-section of the battery module **200** viewed from above. As described above, the cell assembly **100** may be in the form of a stack of a plurality of battery cells **110** and a shock absorbing member **117**. Each of the plurality of battery cells **110** may include the first tab **119***a* (see FIG. **2**C) and the second tab **119***b* (see FIG. **2**C) provided at both end areas of the battery cell **110**.

In addition, the cell assembly **100** may include a busbar assembly **150** into which the first tab **119***a* and the second tab **119***b* are each inserted and electrically connected. Specifically, the busbar assembly **150** may include: a first busbar **1511** electrically connected to the first tab **119***a* and a first busbar frame **1551** supporting the first busbar **1511**; and a second busbar **1551** electrically connected to the second tab **119***b* and a second busbar frame **1552** supporting the second busbar **1512**.
(37) FIG. **2**B shows an enlarged diagram of S**1** part, which is a part of the cross-section of FIG. **2**A. FIG. **2**C shows an example of a cross-section of the battery module viewed from another direction.
(38) Referring to FIGS. **2**B and **2**C, the cell accommodating space **280** in which the cell assembly **100** is accommodated may be subdivided into several spaces. For example, examples of the subdivided space may include: an installation space **281** in which the cell assembly **100** is positioned; a first space **285** provided outside the first busbar **1511** and the first busbar frame **1551**; a second space **286** provided outside the second busbar **1512** and the second busbar frame **1552**; and a cover space **287** provided between an upper portion of the cell assembly **100** and the module cover **211** (see FIG. **1**).
(39) Since the first insulating cover **171** (see FIG. **1**) and the second insulating cover **172** (see FIG. **1**) may be positioned in the first space **285** and the second space **286**, respectively, the first space and the second space may refer to a space excluding the first insulating cover **171** and the second insulating cover **172**.
(40) In addition to the subdivided spaces, an empty space may also be defined between the plurality of battery cells **110** and the busbar assembly **150**. Specifically, an empty space may also be positioned between a side face of the cell body portion **115** where the first tab **119***a* is positioned and the first busbar frame **1551**, and between a side face of the cell body portion **115** where the second tab **119***b* is positioned and a second busbar frame **1552**. This is because at least a portion of the first tab **119***a* and the second tab **119***b* are inserted into the first busbar **1511** and the second busbar **1512** and are electrically connected, so that an empty space may be formed therebetween.
(41) In other words, the battery module **200** may further include a filler space **288** defined between the plurality of battery cells **110** and the busbar assembly **150**.
(42) More specifically, the battery module **200** include a first filler space **2881** and a second filler space **2882** between one side face of each cell body portion **115** of any one of the plurality of battery cells **110** and the first busbar frame **1551** and between the other side face of each cell body portion **115** of any one of the plurality of battery cells **110** and the second busbar frame **1552**, respectively.
(43) One side face of the cell body portion **115** may be a side face on which the first tab **119***a* is positioned, and the other side face of the cell body portion **115** may be a side face on which the second tab **119***b* is positioned.
(44) When the cell assembly **100** is installed in the case **210**, since the length of the first tab **119***a* and the second tab **119***b* along the height direction of the case **210** is shorter than the length of the battery cell **110**, both the first filler space **2881** and the second filler space **2882** may be communicated.
(45) The battery module **200** may include a filler portion **270** that is positioned in the filler space **288** and made of a material having heat-resistance and/or flame-retardance.
(46) Meanwhile, referring to FIG. **2**C, the materials of the filler portions **270** positioned in the first filler space **2881** and the second filler space **2882** may be different from each other. In addition, the length of the first filler space **2881** and the length of the filler space **288** along the direction from the first tab **119***a* to the second tab **119***b*, that is, the X-direction, may be different from each other. This is intended to use a filler portion **270** with a desired function at a desired position for more efficient space utilization.
(47) The battery module **200** may include a heat dissipating portion **295** at a lower portion of the cell assembly **100**, that is, between the cell assembly **100** and the base panel **2194** (see FIG. **1**). The

heat dissipating portion **295** is intended to transfer heat generated from the cell assembly **100** to the outside.
(48) The heat dissipating portion **295** may be positioned below the first filler space **2881** and the second filler space **2882**. This is because a part that requires heat transfer from the battery cell **100** to the outside is the cell body portion **115**.
(49) Referring to FIG. **2**B, when the filler portion **270** is made of a liquid material, the filler portion **270** may also enter between the plurality of stacked battery cells **110** and the buffering member **117**. This is because, even when the plurality of battery cells **110** and the buffering member **117** are in contact with each other without a gap, the filler portion **270** may enter into a gap that is not visible to the naked eye due to a capillary phenomenon.
(50) However, when the filler portion **270** is in a liquid form, considering the viscosity of the filler portion **270**, the amount entering the gap will not be very large. Likewise, even when the filler portion **270** is in a liquid state, the heat dissipating portion **295** and the filler portion **270** may be separated so that they may be distinguished rather than mixed. This is because the density and viscosity of the filler portion **270** and the heat dissipating portion **295** are different from each other.
(51) In addition, in the battery module **200**, based on a direction from the first tab **119***a* to the second tab **119***b*, the length B**1** of the first space **285** may be different from the length L**1** of the first filler space **2881**. Likewise, the length B**2** of the second space **286** may be different from the length L**2** of the second filler space **2882**.
(52) In general, in the battery module **200**, based on a direction from the first tab **119***a* to the second tab **119***b*, the length B**1** of the first space **285** may the same as the length L**1** of the first filler space **2881**. However, not being limited thereto, the length L**1** of the first filler space **2881** may be different from the length L**2** of the second filler space **2882**.
(53) In addition, based on the height direction of the case **210**, the height H**2** of the battery cell **110** may be different from the height of the filler portion **270**. For example, based on the height direction of the case **210**, the height H**2** of the battery cell **110** may be smaller than the height of the filler portion **270**. This is due to the height of the heat dissipating portion **295**.
(54) In other words, referring to FIG. **2**C, based on the bottom face of the cell accommodating space **280**, the maximum height H**1** of the filler portion **270** may be different from the maximum height of the plurality of battery cells **110** positioned in the cell accommodating space **280**.
(55) Preferably, based on the bottom face of the cell accommodating space **280**, the maximum height H**1** of the filler portion **270** will be equal to or greater than the maximum height of the plurality of battery cells **110** positioned in the cell accommodating space **280**. However, not being limited thereto, based on the bottom face of the cell accommodating space **280**, when the maximum height H**1** of the filler portion **270** is provided to be half the maximum height of the battery cell **110** or greater, functions of the filler portion **270** described in the present disclosure may be performed.
(56) FIGS. **3**A and **3**B show another example of a filler portion **270**. The filler portion **270** may include a heat-resistant and/or flame-retardant material. Therefore, as long as the material is a heat-resistant and/or flame-retardant material, the phase of the filler portion **270** may be liquid or solid.
(57) In addition, the filler portion **270** may be a composite material made of several materials rather than a single material. Some materials may be provided to satisfy heat-resistance, and other materials may be provided for flame-retardancy. In addition, rather than being mixed and provided as one integrated form, the various materials may be separated and have a stacked form.
(58) When thermal runaway occurs at a specific battery cell **110** of the cell assembly **100**, the temperature will rise rapidly. As a result, an electrolyte or the like provided inside the battery cell **110** where thermal runaway has occurred may be evaporated to generate a gas, which may cause a swelling phenomenon in the battery cell **110**. When a swelling phenomenon occurs, a joint portion of the tab portion **119** may be gaped.
(59) In other words, as a joint portion of the tab portion **119** is gaped, a high-temperature gas or smoke trapped inside may be discharged in a direction where the tab portion **119** is positioned. This

may be referred to as thermal runaway. This may cause a high-temperature gas or smoke to be discharged in an unintended direction, thereby accelerating heat propagation to other adjacent battery cells **110**.
(60) Therefore, when thermal runaway occurs, the filler portion **270** may prevent a joint portion of the tab portion **119** and the cell body portion **115** (see FIG. **2**C) from gaping, thereby preventing a high-temperature gas or smoke from being discharged in a direction where the tab portion **119** is positioned.
(61) Alternatively, the filler portion **270** may mitigate discharge of the high-temperature gas or smoke in a direction where the tab portion **119** is positioned through a joint portion of the tab portion **119** and the cell body portion **115** (see FIG. **2**C).
(62) Referring to FIG. **3**A, the filler portion **270** may be provided in a solid form. That is, the filler portion **270** may be molded into a desired shape in advance before being assembled into the cell assembly **100**.
(63) Meanwhile, referring to FIG. **3**A, an example is shown in which the buffering member **117** is each provided between battery cells **110** grouped in pairs. This is just one example, and the buffering member **117** does not need to be disposed between the battery cells in an equally set interval.
(64) FIG. **3**B shows an example in which the filler portion **270** is provided in a liquid state and filled in the filler space **288** (see FIG. **2**C). Even when it is in a liquid state, considering the viscosity, which will be described later, the filler portion **270** may remain in the filler space **288** even after time passes.
(65) As described above, the filler portion **270** may include a heat-resistant and/or flame-retardant material. Preferably, the filler portion **270** may be formed of a solid, liquid, or post-cured liquid type material.
(66) For heat-resistance, the filler portion **270** may include a silicone resin, an epoxy resin, or a urethane resin. Preferably, the filler portion **270** may have a 5% weight loss temperature of 400° C. or higher in a thermogravimetric analysis (TGA), or an 800° C. remaining weight of 70% by weight or higher. Due to these characteristics, the stability of the battery module at high temperatures can be further improved. In the thermogravimetric analysis (TGA), measurement may be performed within a range of 25° C. to 800° C. at a temperature increase rate of 20° C./min in a nitrogen (N.sub.2) environment of 60 cm.sup.3/min.
(67) In addition, the filler portion **270** may also perform a function of securing the plurality of battery cells **110**. For example, a solid state filler portion **270** or a filler portion **270**, which is injected in a liquid state and then cured, may prevent the tab portion **119** from being deformed by an external shock or the like, after the tab portion **119** is inserted into the busbar assembly **150**. A liquid state filler portion **270** will also be able to prevent deformation of the tab portion **119** to some extent due to the viscosity thereof.
(68) In particular, a filler portion **270**, which is injected in a liquid state and then cured, specifically includes an acrylic resin, an epoxy resin, a urethane resin, an olefin resin, an ethylene vinyl acetate (EVA) resin, a silicone resin, or the like.
(69) As described above, for the functions of the filler portion **270**, various materials may be mixed. In addition, a liquid state filler portion **270** or a cured type filler portion **270**, may further include a viscosity modifying agent, for example, a thixotropic agent, a diluent, a dispersant, a face treatment agent, or a coupling agent, in order to control the viscosity in a liquid state.
(70) The thixotropic agent may adjust the viscosity of a resin composition according to the shear force to ensure that the manufacturing process of the battery module is carried out effectively. Examples of applicable thixotropic agents include fumed silica or the like. Preferably, the viscosity of the filler portion **270** may be 100 cPs (centi-poise) or more and 600 (cPs) or less based on 25° C.
(71) A diluent or a dispersant is usually used to lower the viscosity of a resin composition, and various types known in the industry may be used without limitation as long as they can exhibit the

abovementioned effect. A face treatment agent is for face treatment of a filler introduced into the filler portion **270**, and various types known in the industry can be used without limitation as long as they can exhibit the abovementioned effect.
(72) The filler portion **270** may include a flame-retardant material. A flame-retardant material refers to a material that is difficult to melt at high temperatures. For example, various known flame-retardants may be applied as a flame-retardant material without any particular limitation, and for example, flame-retardants in the form of a solid filler or liquid flame-retardants may be applied. Flame-retardants include, for example, organic flame-retardants such as melamine cyanurate and inorganic flame-retardants such as magnesium hydroxide, but are not limited thereto.
(73) For example, in the present specification, a flame-retardant material may mean a polymer material that has a V-0 class in the 94V test (vertical burning test) of Underwriter's Laboratory (UL), which is a flame-retardant standard for polymer materials.
(74) Specifically, the filler portion **270** may further include a heat-resistant or flame-retardant material. The heat-resistant or flame-retardant material include phosphorus-based, halogen-based, and inorganic flame-retardants, and a phosphorus-based flame-retardant material may preferably include a phosphate compound, a phosphonate compound, a phosphinate compound, a phosphine oxide compound, a phosphazene compound, and a metal salt thereof. These may be used individually or in combination of two or more types.
(75) Preferably, a temperature range in which the filler portion **270** has the intended heat resistance and flame-retardancy may be −60° C. or higher and 120° C. or lower.
(76) In addition, the solid or cured filler portion **270** may preferably have appropriate hardness. That is, this is to ensure the impact resistance and vibration resistance of the filler portion **270** without high brittleness of the filler portion **270** so that the filler portion **270** has improved durability. Preferably, the hardness of the filler portion **270** may be 35 or more and 45 or less.
(77) In addition, the filler portion **270** may include an electrically insulating material. Through this, the dielectric strength of the pillar portion **270** may preferably be at least 3.0 (kV/mm).
(78) In addition, the thermal conductivity of the filler portion **270** may preferably be 0.05 (W/m-K) or less. Considering that the filler portion **270** is a heat-resistant material, the thermal conductivity of the filler portion **270** will be smaller than that of the heat dissipating portion **295**.
(79) When the filler portion **270** is a foam material, which is injected in a liquid state and then cured, considering the thermal conductivity and use temperature range required for the filler portion **270**, the foam density of the foam material may be 0.1 (g/cm.sup.3) or more and 0.2 (g/cm.sup.3) or less upon free rise.
(80) FIG. **4**A shows an example of a filler portion **270** molded into a preset shape; and FIG. **4**B shows an enlarged diagram of a part of the filler portion **270**.
(81) Referring to FIG. **4**A, the filler portion **270** may be in a solid form. That is, the filler portion **270** may be molded into a preset shape before being coupled to the cell assembly **100**. Preferably, the filler portion **270** may be provided in the form of a foam.
(82) The filler portion **270** may be coupled after inverting the battery module **200** being assembled. In the conventional process, in the cell assembly **100** (see FIG. **1**), the module cover **211** may be coupled before the module body **219** (see FIG. **1**). Therefore, to insert the filler portion **270** between the plurality of battery cells **110** and the busbar assembly **150** in a state in which the module cover **211** is coupled, the battery module **200** being assembled will be inverted, and then the filler portion **270** will be inserted.
(83) Specifically, the filler portion **270** may include: a first filler portion **2701** accommodated in the first filler space **2881** (see FIG. **2**C); and a second filler portion **2702** accommodated in the second filler space **2882** (see FIG. **2**C).
(84) FIG. **4**B is an enlarged view of S**2** region in FIG. **4**A. For explanation, a direction toward the cell assembly **100** (see FIG. **1**) is determined as front (F), and a direction toward the plurality of battery cells **110** (see FIG. **1**) is indicated as rear (R). In addition, referring to FIG. **4**A, the Z-

direction is indicated downward, which is intended to emphasize that the module cover **211** is below the cell assembly **100**, as the battery module **200** being assembled is inverted.

(85) In order to be accommodated between the plurality of battery cells **110** and the busbar assembly **150**, the filler portion **270** may be provided in an individual form to be inserted between tab portions **119** of each of any one battery cell **110** (see FIG. **1**) and another adjacent battery cell **110**. However, in this case, the assembly time may be long, and a lot of empty space may remain between the plurality of battery cells **110** and the busbar assembly **150**.

(86) To this end, the filler portion **270** may further include: a body portion **2701***b* inserted between tab portions **119** of each of the plurality of battery cells **110**; and a connecting portion **2701***c* positioned at one end positioned farther from the module cover **211**, among both ends of the body portion **2701***b*, and connecting the body portion **2701***b*.

(87) When the body portions **2701***b* are provided in a plural number, in order that the filler portion **270** may not interfere with electrical connection of each of the tab portions **119** to the busbar assembly **150**, the plurality of body portions **2701***b* may be spaced in a constant interval and further include inserting portions **2701***a* in the form of a hole or a groove between the plurality of body portions. The positions of the insertion portions **2701***a* will be arranged to correspond to the positions of the tab portions **119**.

(88) When the filler portion **270** is positioned in the filler space **288** (see FIG. **2**C), the connecting portion **2701***c* is caught on the tab portion **119** and thus it may play the role of a stopper upon the installation of the filler portion **270**.

(89) In addition, the connecting portion **2701***c* may be provided in a form extending toward the rear where the busbar assembly **150** is positioned. This is intended to fill a space between the busbar assembly **150** and the base panel **2194** (see FIG. **1**).

(90) FIG. **5** shows an example of a method of injecting a filler portion **270** injected in a liquid form. Referring to FIG. **5**, the Z-direction is indicated downward, which is intended to emphasize that the module cover **211** is below the cell assembly **100**, as the battery module **200** being assembled is inverted.

(91) The battery module **200** being assembled, shown in FIG. **5**, has an end portion **180** and an insulating cover **170** coupled, and after the module cover **211** is coupled, the battery module **200** being assembled is in an inverted form. Therefore, it can be seen that the end portion **180** and the insulating cover **170** play the role of side faces, and the module cover **211** plays the role of a bottom face so that a type container that can contain a liquid is formed by the module cover **211**, the end portion **180**, and the insulating cover **170**.

(92) Therefore, the inverted battery module **200** may accommodate a liquid filler portion **270** in a filler space **288**. FIG. **5** shows an example of an injecting portion **900** injecting a filler portion **270** of a liquid form into a first filler space **2881** and a second filler space **2882**.

(93) The injecting portion **900** may include dispensers each disposed between a tab portion **119** (see FIG. **2**C) and another tab portion **119** of a plurality of battery cells **110**. Through this, the injection portion **900** will be able to inject a liquid filler portion **270** into all filler spaces **288** simultaneously. For understanding, FIG. **5** does not show all of the injection portions **900**, but only shows them sparsely in a few places.

(94) In contrast, the injection portions **900** may be disposed one by one in the first filler spaces **2881** (see FIG. **2**C) and the second filler spaces **2882** (FIG. **2**C) to sequentially fill each filler space **288** with a liquid filler portion **270**. However, considering the time required to fill the filler spaces **288** with a filler portion **270**, it may be preferable to fill the filler spaces **288** simultaneously. FIGS. **6**A and **6**B briefly show an example of a method of assembling the battery module **200**. FIG. **7** shows a flowchart of an example of the method of assembling the battery module **200**.

(95) Referring to FIGS. **6**A and **7**, the assembling method of the present disclosure includes a step S**100** of assembling a cell assembly **100** including the plurality of battery cells **110** and a busbar assembly **150** electrically connected to the plurality of battery cells **110**.

(96) Specifically, referring to FIGS. **6**A and **7**, the step S**100** of assembling a cell assembly **100** may include a stacking step S**110** of stacking the plurality of battery cells **110** and a buffering member **117** positioned between the plurality of battery cells **110**. In addition, the stacking step S**110** may include stacking each of a first end plate **181** and a second end plate **182** at the first and last in a perpendicular direction of the busbar assembly **150** or in a direction in which the plurality of battery cells **110** are stacked.
(97) Referring to FIGS. **6**B and **7**, the step S**100** of assembling a cell assembly **100** may further include a step S**150** of connecting a busbar assembly **150** to the stacked battery cells **110**, after the stacking step S**110**.
(98) Since the tab portions **119** of the plurality of battery cells **110** are positioned in an opposite direction with each other, the busbar assembly **150** may also include: a first busbar **1511** and a first busbar frame **1551** coupled to one face to which the first tab **119***a* (see FIG. **2**C) is coupled; and a second busbar **1512** and a second busbar frame **1552** coupled to one face to which the second tab **119***b* (see FIG. **2**C) is coupled. The first tab **119***a* may be electrically connected to the first busbar **1511**, and the second tab **119***b* may be electrically connected to the second busbar **1512**.
(99) Referring to FIGS. **6**C and **7**, thereafter, the assembling method of the present disclosure may further include a step S**210** of coupling an insulating cover **170** provided in a parallel direction to the busbar assembly **150** to the cell assembly **100**.
(100) A first insulating cover **171** among the insulating covers **170** may be coupled in a direction in which the first busbar **1511** is positioned, and a second insulating cover **172** among the insulating covers **170** may be coupled in a direction in which the second busbar **1512** is positioned. The first insulating cover **171** and the second insulating cover **172** will be coupled to the first end plate **181** and the second end plate **182**.
(101) Thereafter, referring to FIGS. **6**D and **7**, after coupling the first insulating cover **171** and the second insulating cover **172**, the assembling method of the present disclosure may include a step S**250** of coupling the module cover **211** to the cell assembly **100**. The reason for assembling the module cover **211** before the module body **219** (see FIG. **6**F) is to protect the busbar assembly **150** during the assembly process.
(102) Thereafter, the assembling method of the present disclosure may go through a first inverting step S**300** of inverting the battery module **200** being assembled. The reason for inverting the battery module **200** being assembled is to locate the filler portion **270** into the first filler space **2881** (see FIG. **2**C) and the second filler space **2882** (see FIG. **2**C). Since it is difficult to insert the filler portion **270** from above due to the already assembled module cover **211**, the battery module **200** being assembled is inverted to locate the filler portion **270**.
(103) Referring to FIGS. **6**E and **7**, the assembling method of the present disclosure may include a step S**400** of forming the filler portion **270** after the first inverting step S**300**.
(104) When the filler portion **270** has a preset shape in a solid state, like an example shown in FIG. **6**E, the filler portion **270** may be caused to approach the module cover **211** from an opposite direction of the module cover **211** to locate the filler portion **270** to the filler space **288**.
(105) When the filler portion **270** is in a liquid state or a type that is injected as a liquid and then cured, like an example shown in FIG. **5**, the filler portion **270** may be formed by filling the same into the filler space **288** using the injection portion **900**.
(106) Referring to FIGS. **6**F and **7**, the assembling method of the present disclosure may further include a step S**550** of coupling the module body **219** to the module cover **211**.
(107) Since the battery module **200** being assembled has already been inverted, the module body **219** may be caused to approach the module cover **211** in a direction from top to bottom for coupling S**550** to the module cover **211**.
(108) Meanwhile, as described above, the battery module **200** may further include a heat dissipating portion **295** on a base panel **2194** (see FIG. **1**), which is a lower face of the module body **219**, or on a bottom face of the cell accommodating space **280**. To this end, the assembling

method of the present disclosure may include a step of forming the heat dissipating portion **295** on the base panel **2194** by injecting or coating the heat dissipating portion before coupling S**550** the module body **219** to the module cover **211**.
(109) After the module body **219** and the module cover **211** are coupled, the assembling method of the present disclosure may include a second inverting step S**600** of re-inverting the battery module **200** so that the module cover **211** may be positioned on top. This is for inspecting S**700** of the battery module **200** in a normal state.
(110) The present disclosure may be modified and implemented in various forms, and its scope is not limited to the embodiments described above. Therefore, if a modified embodiment includes components of the present disclosure, it should be regarded as falling within the scope of the rights of the present disclosure.

Claims

1. A battery module comprising: a cell assembly including a plurality of battery cells, one or more of buffering members provided between the plurality of battery cells, and a busbar assembly for electrically connecting the plurality of battery cells, wherein the busbar assembly includes a busbar and a busbar frame supporting the busbar; a case defining a cell accommodating space for accommodating the cell assembly therein; and a filler portion provided between the plurality of battery cells and the busbar frame or between one or more of buffering members and the busbar frame inside the case and including a flame-retardant material.
2. The battery module according to claim 1, wherein the plurality of battery cells each include: a cell body portion generating or storing electrical energy; and a tab portion protruding from the cell body portion toward the busbar assembly and electrically connecting the cell body portion and the busbar assembly, and wherein the filler portion is positioned in a filler space defined between one side face of the cell body portion from which the tab portion protrudes and the busbar assembly.
3. The battery module according to claim 2, wherein the tab portion includes a first tab and a second tab protruding from the cell body portion toward the busbar assembly along any one direction among directions perpendicular to the height direction of the case, wherein the busbar is formed by a plurality of busbars, and the plurality of busbars include: a first busbar electrically connected to the first tab and a second busbar electrically connected to the second tab, wherein the busbar frame is formed by a plurality of busbar frames, and the plurality of busbar frames include a first busbar frame supporting the first busbar and a second busbar frame supporting the second busbar, and wherein the filler space includes: a first filler space provided between a part in which the first tab is positioned and the first busbar frame; and a second filler space provided between a part in which the second tab is positioned and the second busbar frame.
4. The battery module according to claim 3, wherein, along a direction from the first filler space to the second filler space, the length of the first filler space is different from the length of the second filler space.
5. The battery module according to claim 3, further comprising: a heat dissipating portion positioned between the cell assembly and a bottom face of the cell accommodating space to dissipate heat generated from the plurality of battery cells to the outside through the case, wherein the heat dissipating portion is positioned between the first filler space and the second filler space.
6. The battery module according to claim 2, wherein the filler portion is formed of a polymer material that is cured after being injected into the filler space.
7. The battery module according to claim 2, wherein the filler portion is filled in the filler space in a liquid form.
8. The battery module according to claim 2, wherein the filler portion includes: body portions inserted between tab portions of the plurality of battery cells; and a connecting portion positioned lower than the body portions, the connecting portion configured to connect each end of the body

portions.
9. The battery module according to claim 3, further comprising: a first insulation cover positioned in a direction from the outside of the first busbar frame to one side face of the case which is positioned closer to the first busbar frame, among both side faces facing the first busbar frame, to insulate between the first busbar and the case; and a second insulation cover positioned in a direction from the second busbar frame to the other face of the case to insulate between the second busbar and the case.
10. The battery module according to claim 9, wherein the cell assembly further includes a first end plate and a second end plate respectively connected to the first insulation cover and the second insulation cover to surround the plurality of battery cells.
11. The battery module according to claim 10, further comprising: a module cover coupled to the first insulation cover, the second insulation cover, the first end plate, and the second end plate.
12. The battery module according to claim 1, wherein, based on a bottom face of the cell accommodating space, the maximum height of the filler portion is more than half of the maximum height of the plurality of battery cells positioned in the cell accommodating space.
13. The battery module according to claim 1, wherein the case includes: a module body including an opening portion in an upper portion thereof, extending along a first direction, which is any one direction among directions perpendicular to the height direction of the case, and formed in a U-shape of which both side faces perpendicular to the first direction are open; and a module cover coupled to the module body to close the opening portion.
14. The battery module according to claim 13, further comprising: a heat dissipating portion positioned between the cell assembly and a bottom face of the cell accommodating space to dissipate heat generated from the plurality of battery cells to the outside through the case.
15. The battery module according to claim 1, wherein the filler portion further includes a heat-resistant material.